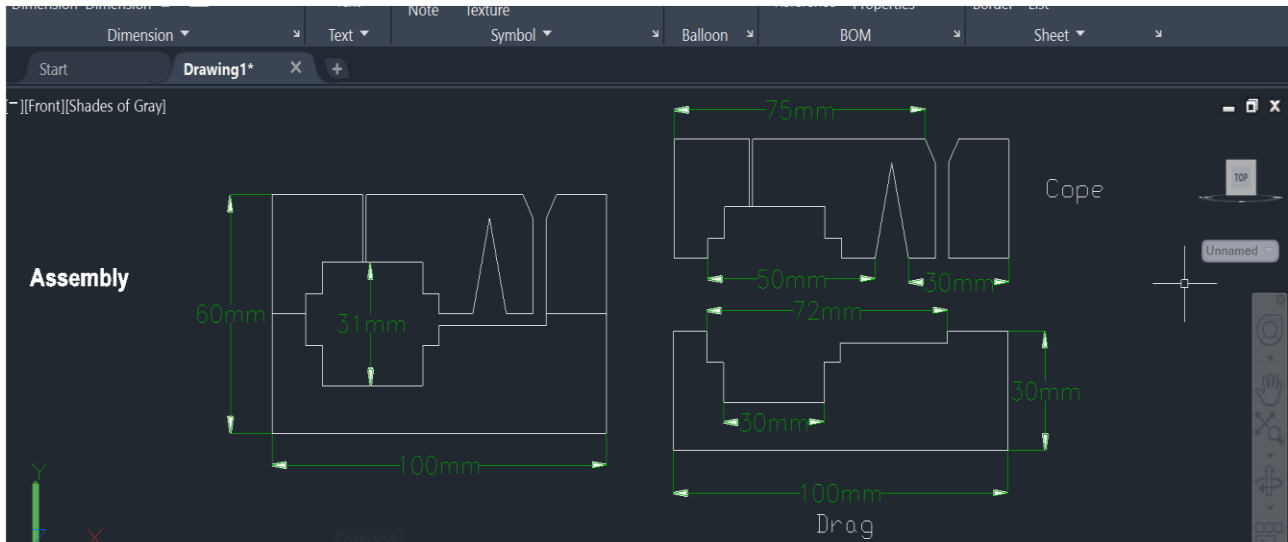
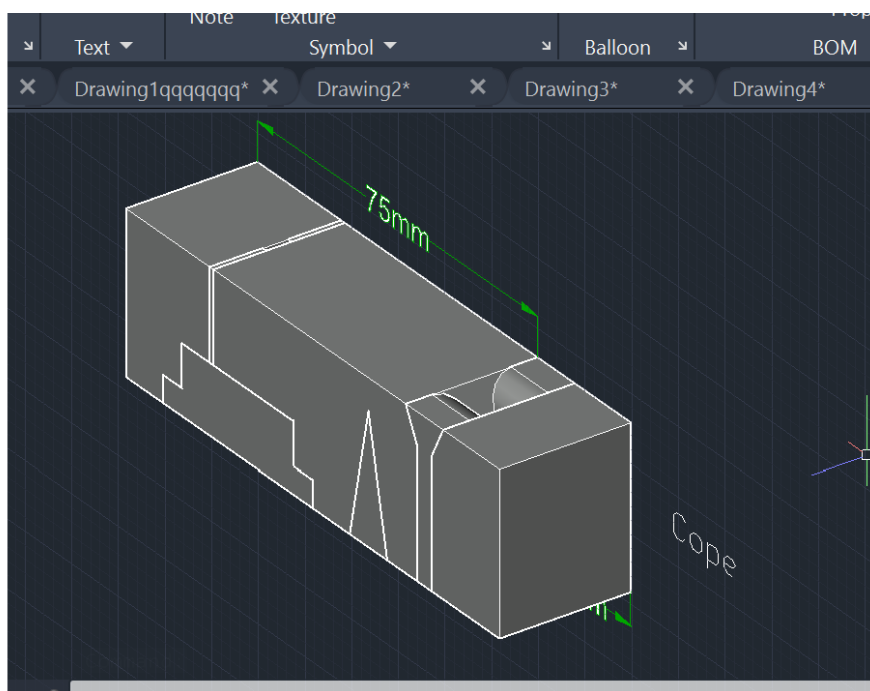


Sand Mold Design Report(2D View):

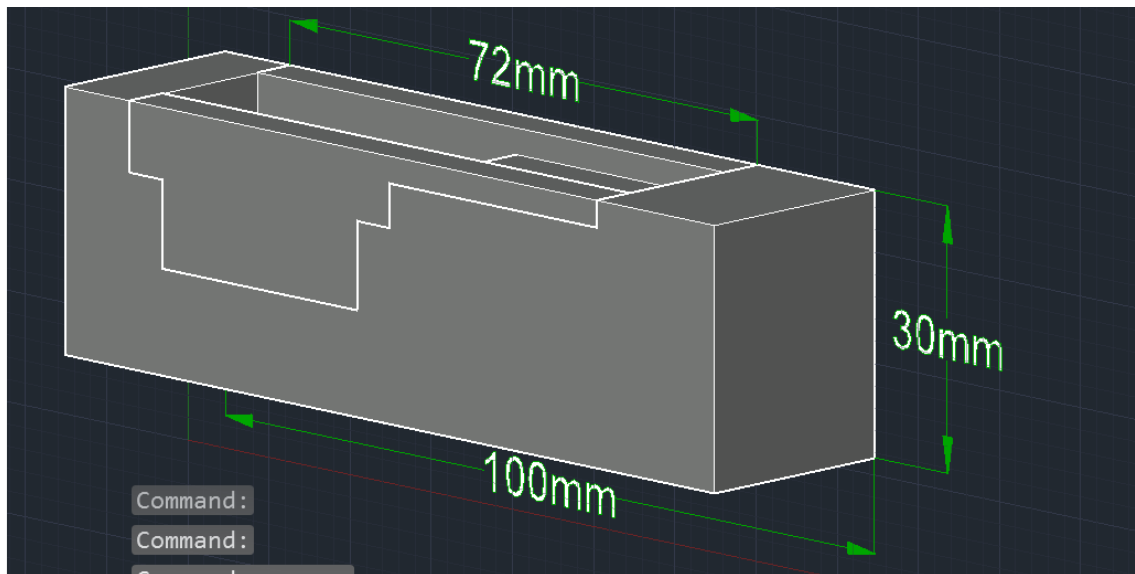


Object and Mold Drawings

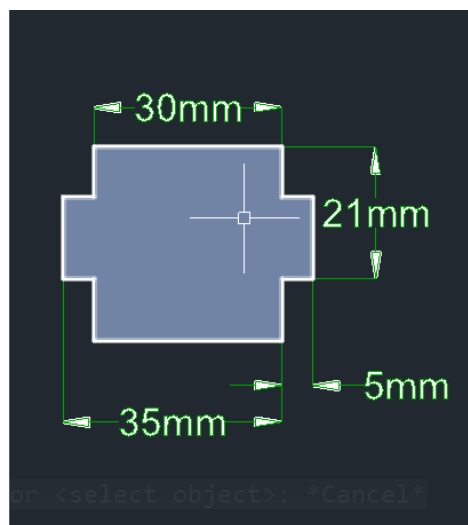
Cope: Upper mold half showing riser, sprue, and alignment features.



Drag: Includes lower mold half showing cavity, runner, gate, and support structure.



Modified rectangular box Mould:



Material Chosen: Aluminum

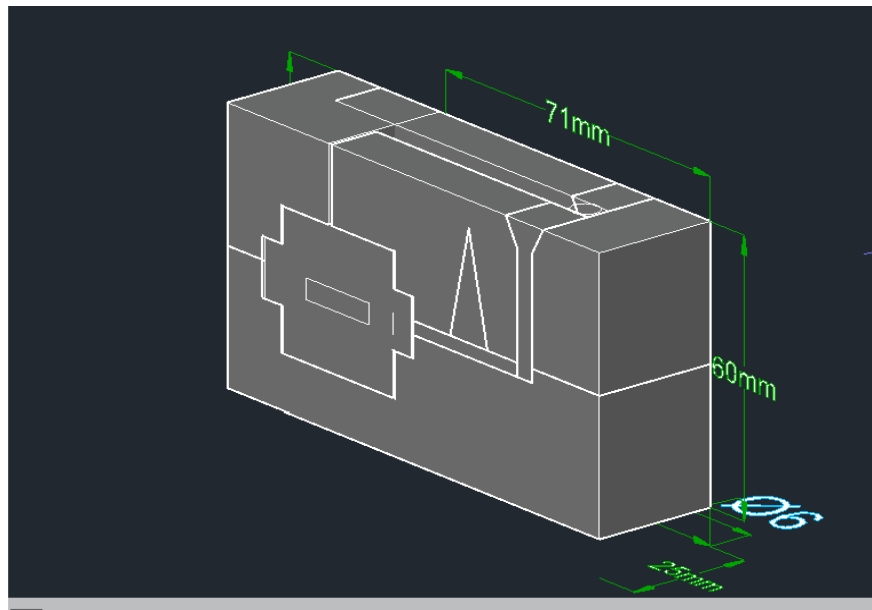
• **Tolerances:**

- Linear: ± 0.5 mm

Angular: $\pm 1^\circ$

• **Drawings Include:**

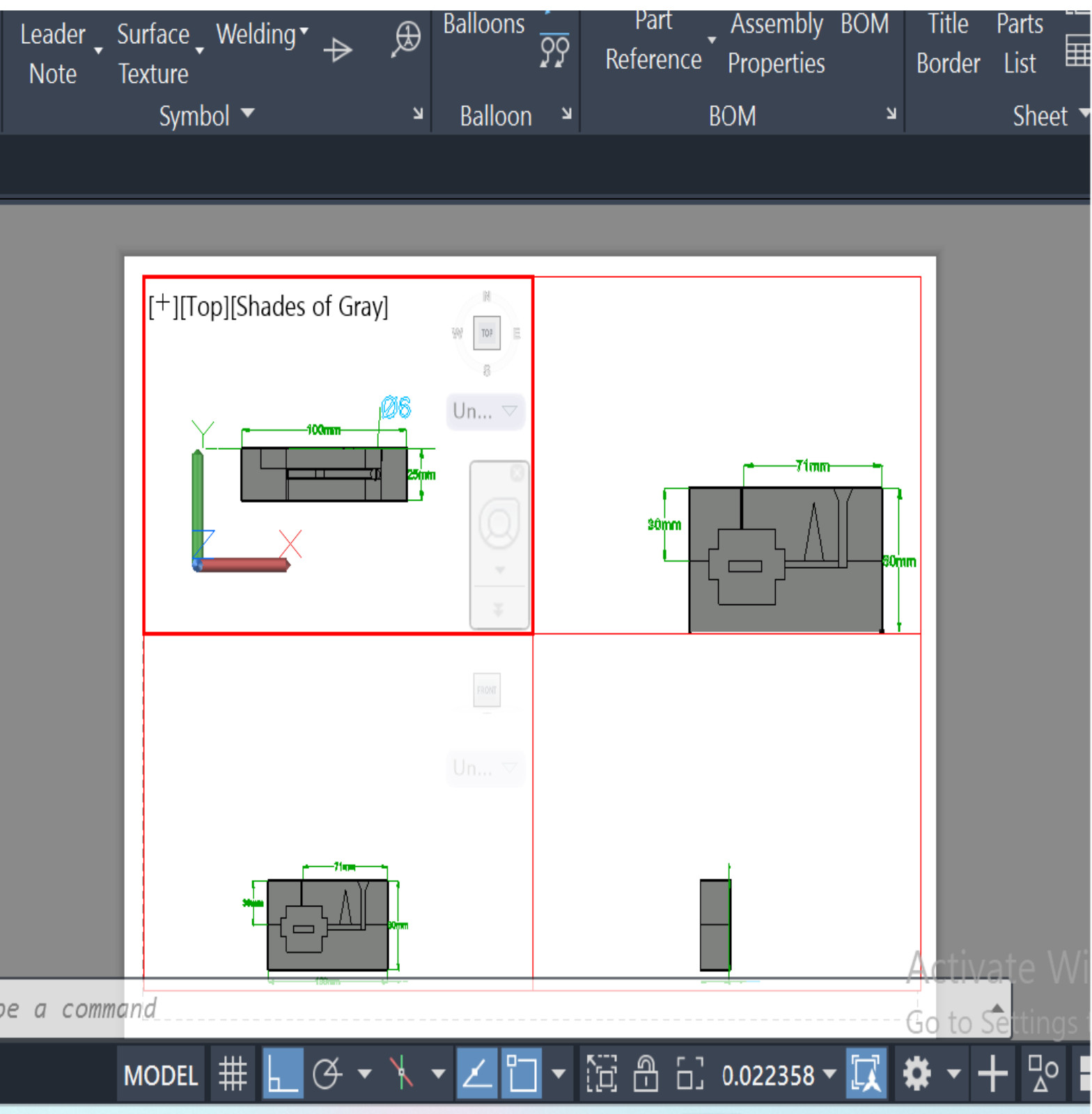
- Front, Top, and Side Views of the Object with complete dimensions
- Front, Top, and Side Views of the Mold (Cavity) with dimensions
- Combined view showing both Drag and Cope aligned.
- All mold components labeled: Pattern, Core (if used), Sprue, Riser, Gate, Runner, Cope, and Drag.



Design Considerations Used:

- Draft angle provided for easy pattern removal (typically 1° – 3°)
- Rounded internal corners to reduce stress concentration
- Uniform wall thickness to avoid hot spots and defects
- Proper riser placement to avoid shrinkage

All views front,top, side of assembly:



The sand mold design successfully meets the requirements for casting the chosen 3D object with proper dimensions, tolerances, and allowances. Both cope and drag sections are clearly defined with labeled components, ensuring accurate mold assembly

